



Educator

Training

Module 2: Water Safety & Health

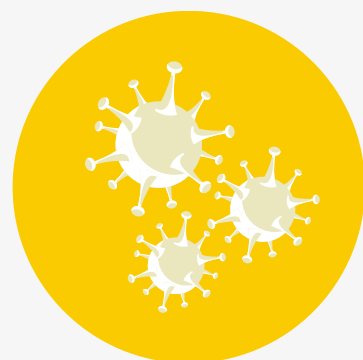
Version 2.0 (March 2026)

WASH-Ed Learning Curriculum



Module 1:

Water Resources & Accessibility



Module 2:

Water Safety & Health



Module 3:

Water Sustainability



Module 4:

Sanitation



Module 5:

Hand Hygiene



Module 6:

Disinfection & Other Hygienic Practices



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Our Learning

Objective

- 💧 **Identify** various water-related diseases and how they are transmitted
- 💧 **Describe** pathogens and the contamination of water resources
- 💧 **Discuss** the importance of a reliable and safe water supply
- 💧 **Explain** the effects of climate change and disasters on water safety
- 💧 **Propose** tips to your community on how to avoid water-related diseases, especially after a disease

What We

Will Talk About



1	Water-related Diseases
2	Pathogens
3	Effects of Climate Change and Disasters
4	Checking for Understanding
5	Activities to teach Module 2

Topic 1

Water-related diseases

Understand different water-related diseases and their transmission methods to recognize the importance of good hygiene practices in preventing these illnesses

01

Waterborne Diseases

What are waterborne diseases and how are they transmitted?

02

Other Water-related Diseases

What are the other types of diseases and how are they different?

03

Importance of Hand Hygiene

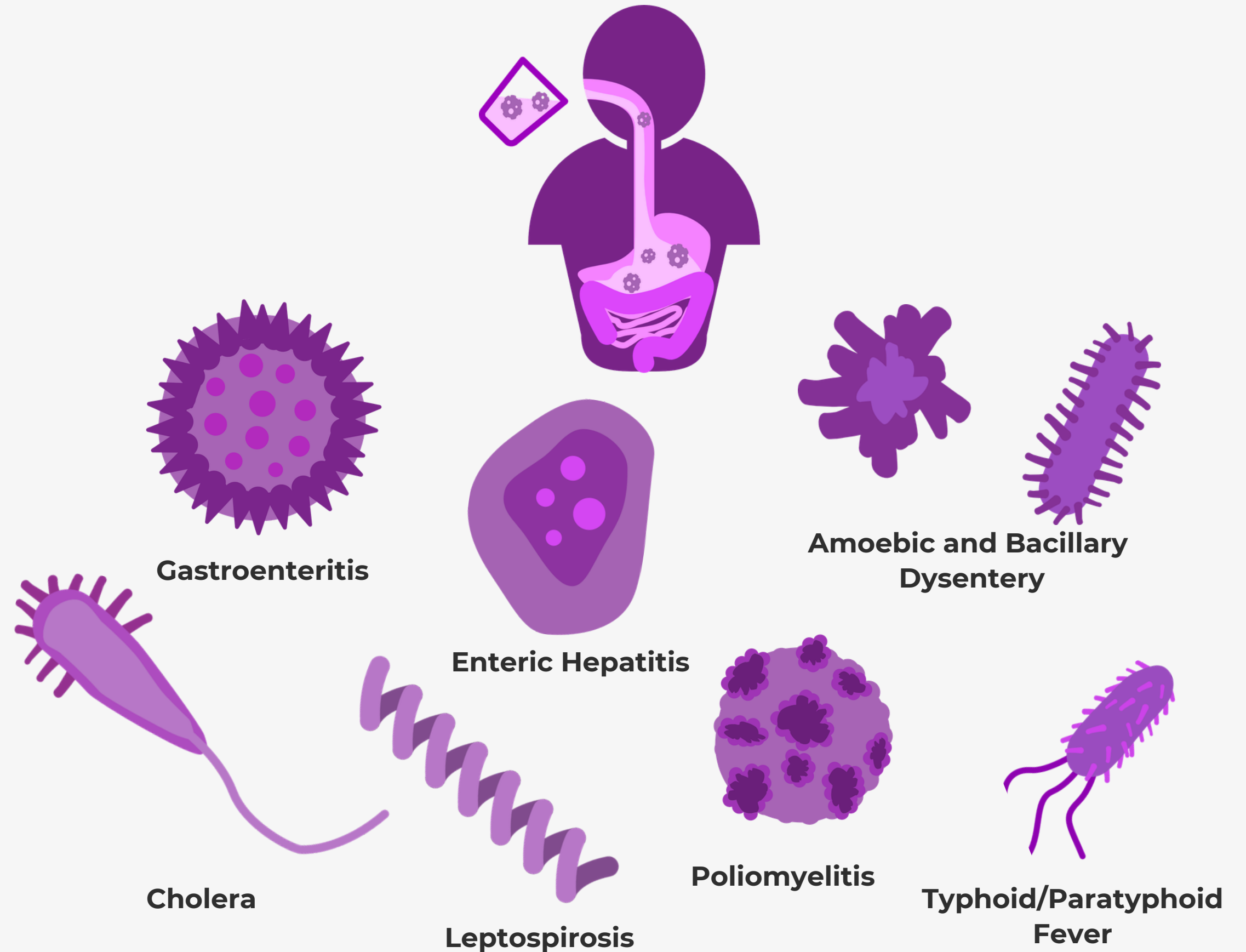
What role do hands play in protecting us from these illnesses?

Waterborne Diseases



Makes up ~80% of water-related diseases

Transmitted through the **consumption** of water contaminated by human or animal feces that contain **pathogenic** bacteria, viruses, or parasites



Other Water-related Diseases

The other four groups make up the remaining 20%

1

Water-washed
Diseases

2

Water-aerosol
Diseases

3

Water-based
Diseases

4

Water-related
arthropod vectors



Can you guess how each of these other diseases is transmitted?

Other Water-related Diseases

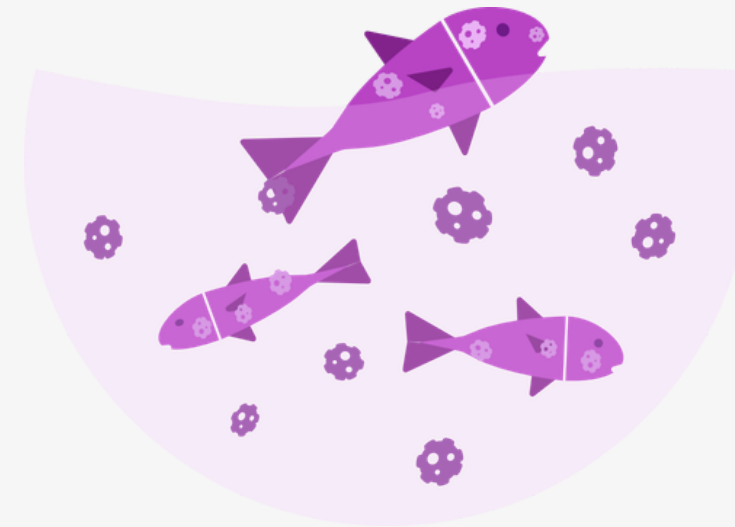
The other four groups make up the remaining 20%



**Water-washed
Diseases**



**Water-aerosol
Diseases**



**Water-based
Diseases**

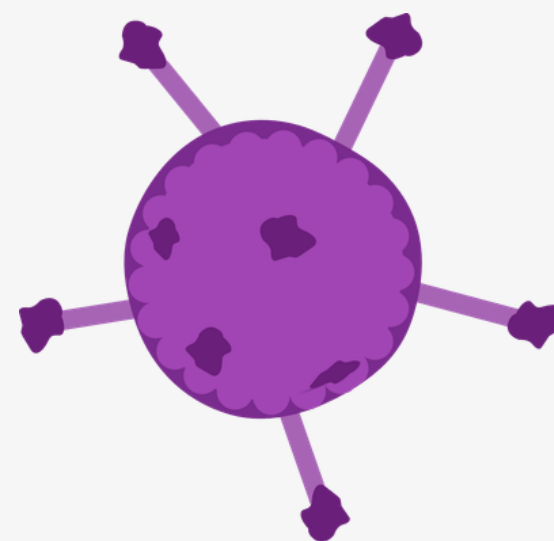
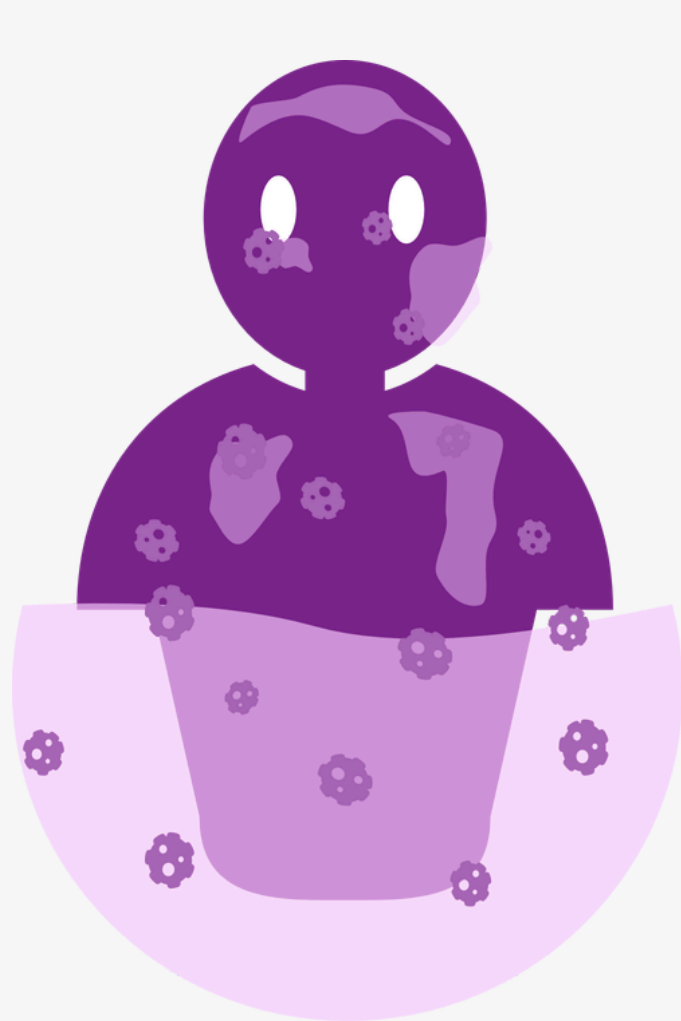


**Water-related
arthropod vectors**

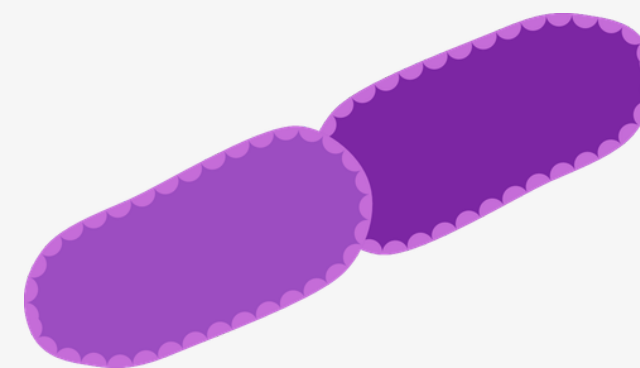
Other Water-related Diseases

Water-washed diseases

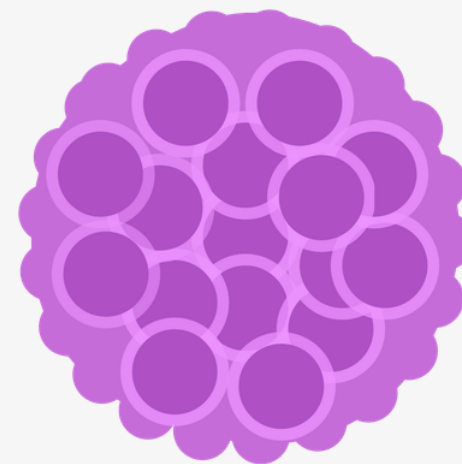
Transmitted through your skin, ears, and eyes making contact with contaminated water and personal poor hygiene



Conjunctivitis



Leprosy



Trachoma



Scabies

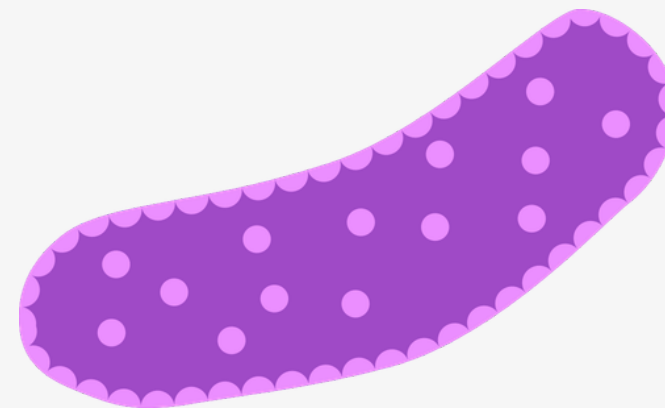
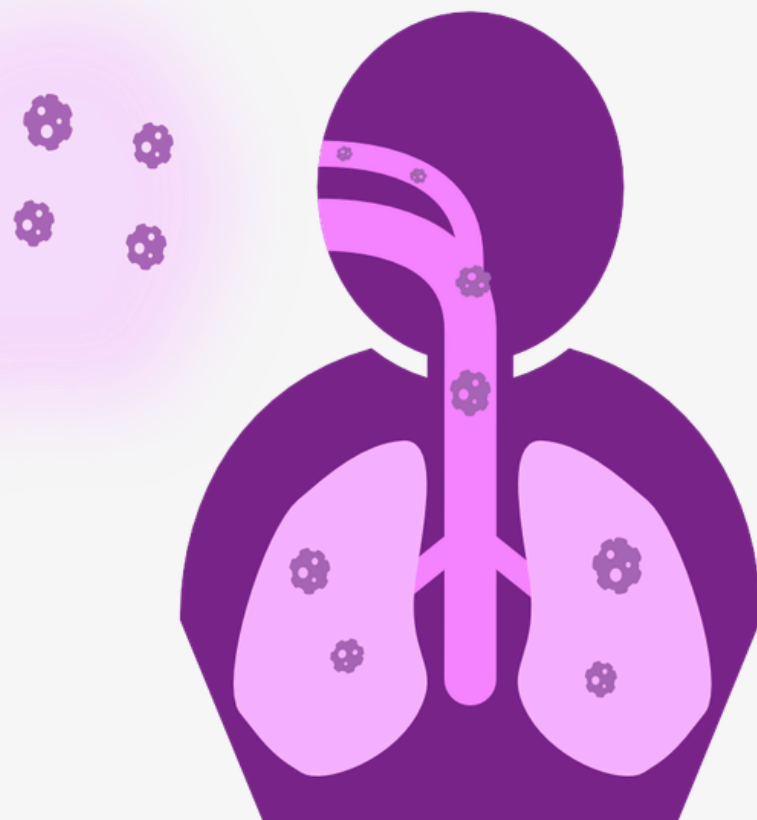


Intestinal helminth
Infections

Other Water-related Diseases

Water-aerosol diseases

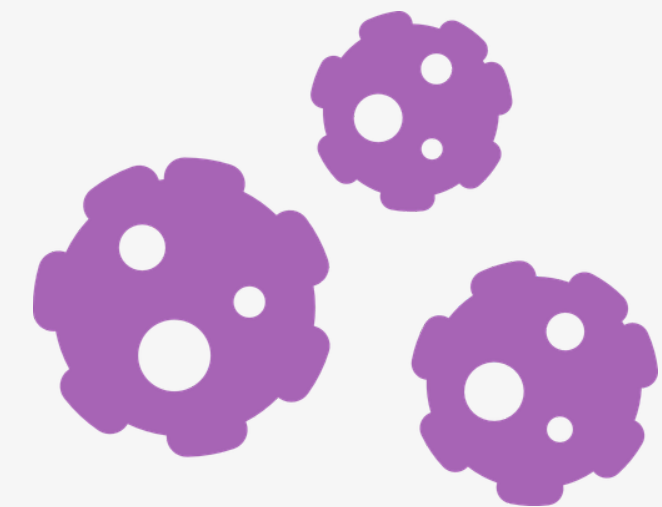
Transmitted through the inhalation of water aerosols containing pathogens



Legionellosis



Pfiesteria

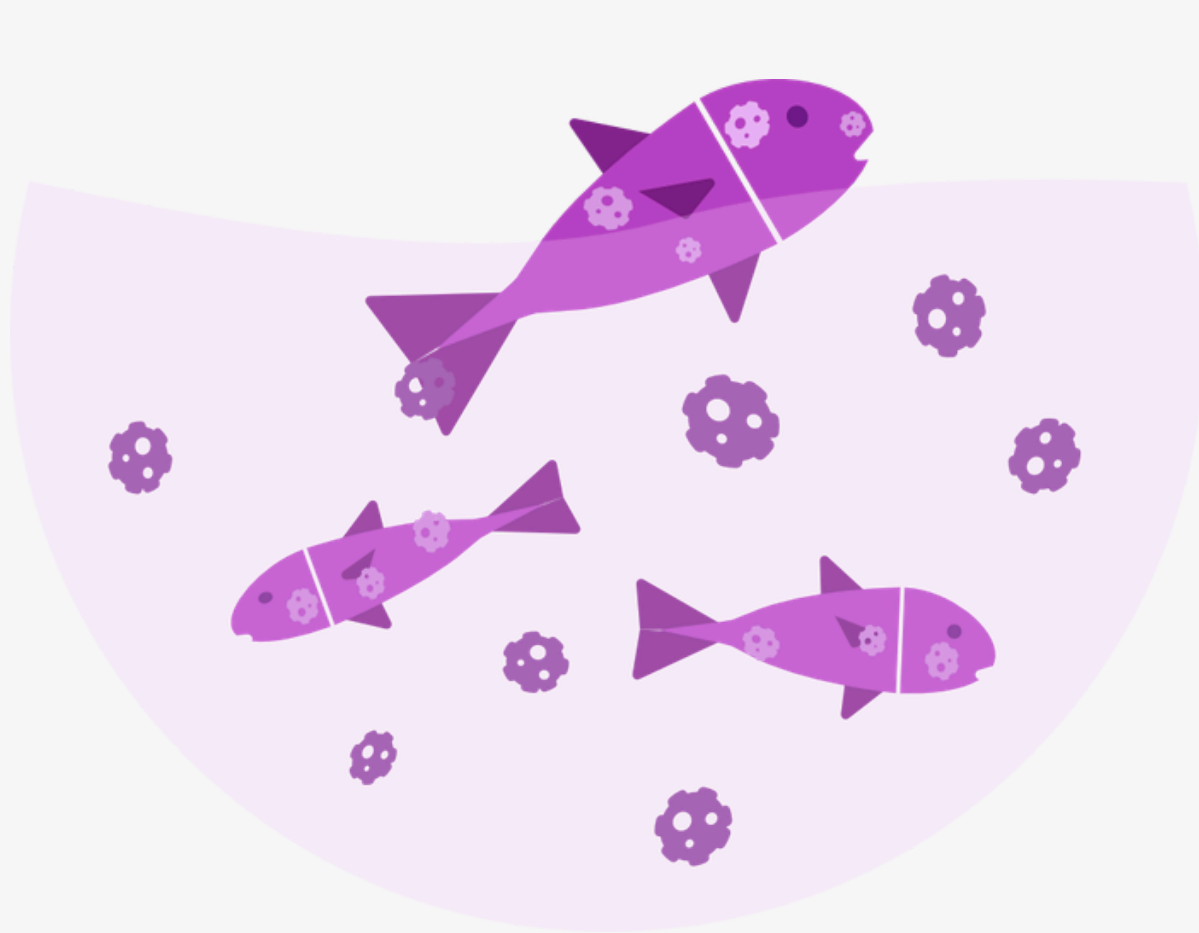


Coronavirus

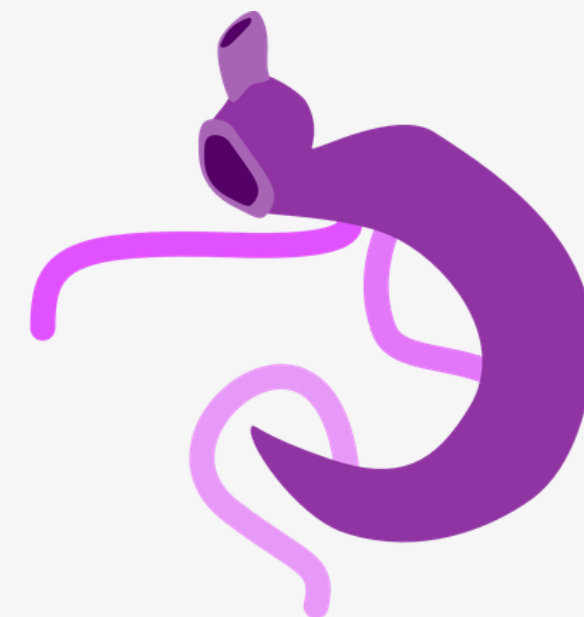
Other Water-related Diseases

Water-based diseases

Transmitted through parasites found in intermediate organisms that can be found in contaminated water



Dracunculiasis

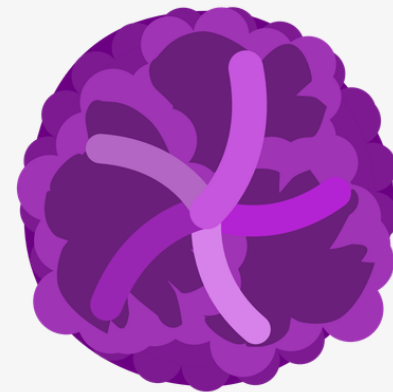


**Schistosomiasis
(Bilharziasis)**

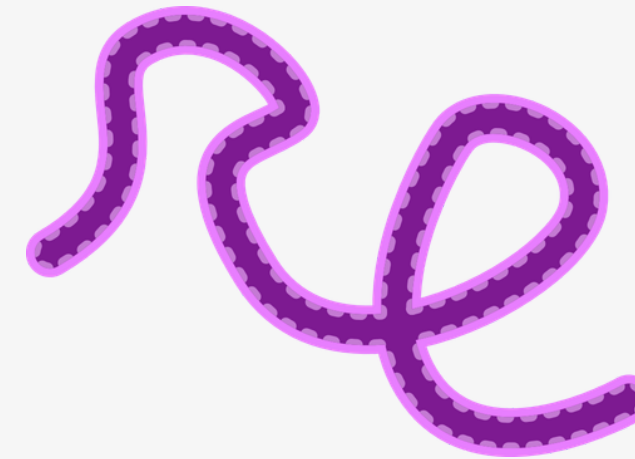
Other Water-related Diseases

Water-related arthropod vectors

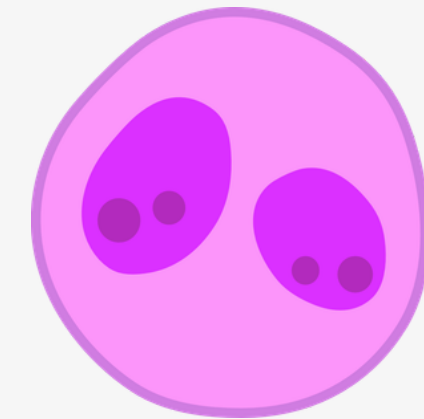
Transmitted through insect vectors breeding in contaminated water



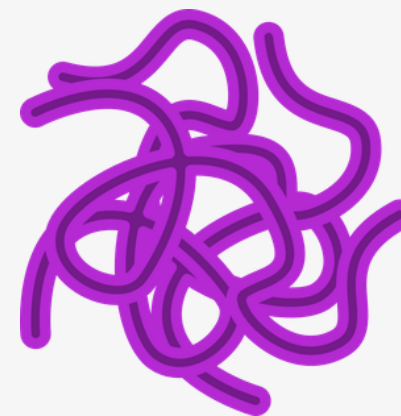
Dengue



Lymphatic Filariasis



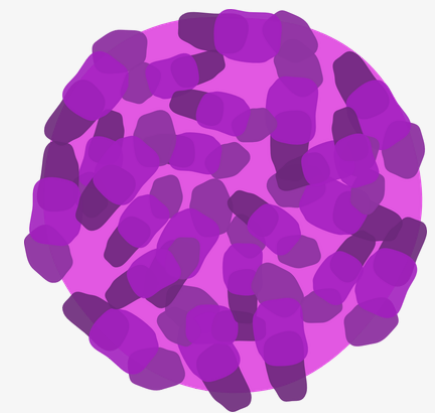
Malaria



Onchocerciasis



Trypanosomiasis



Yellow Fever



Importance of Hand Hygiene

Germs on your hands can get inside your body through:

- 💧 eating (**consumption**)
- 💧 bodily contact (**touching your eyes, nose, or mouth**)
- 💧 poor hand hygiene which puts both ourselves and others at risk

Good hand hygiene practices help us avoid contracting illnesses from water-related diseases



This will be explored further in Module 5

Topic 2 Pathogens

Understand the types of pathogenic microorganisms, their impact on health, and the importance of safe water practices to prevent water-related diseases

01

Introduction to Pathogens

What are pathogens?

02

Water Contamination

How might water be contaminated by pathogens and other substances?

03

Impact on Communities

What can we do to help make our communities more safe?

Introduction to Pathogens

Trillions of microbes are living in and on your body right now. They help you digest food, protect you against infections, and even maintain your reproductive health.

However, some microorganisms are **harmful** and can make you ill. These are referred to as **pathogens** which can be found in **contaminated water**.

Bathing, washing, drinking water, or eating food exposed to contaminated water spreads waterborne diseases.



How might water be contaminated by pathogens?



Water Contamination

How might water be contaminated
by pathogens?

2 main ways

1



Water sources not being
properly treated before
consuming

2



Practice of open
defecation

Water Contamination

Other contaminants come from:

- 💧 **heavy metals**
- 💧 **toxic agents**
- 💧 **persistent organic pollutants**
- 💧 **inorganic chemicals (e.g. inorganic arsenic compounds)**

In the Philippines, there are numerous reported cases of **arsenic poisoning** from contaminated groundwater.

Arsenic poisoning is a chronic condition brought on by consuming water with high arsenic levels over an extended period of time.



Impact on Communities



What actions can we take?

- 1** Ensure your school has access to safe water supplies
- 2** Educate young learners so they are more aware of the water sources they use
- 3** Keep school administrations, water service providers, and local authorities accountable

Topic 3

Effects of Climate Change and Disasters

Understand the effects of climate change and disasters on water safety and health, and practical measures to prevent water-related diseases in the aftermath of natural disasters

01

Effects of climate change

Point Source

02

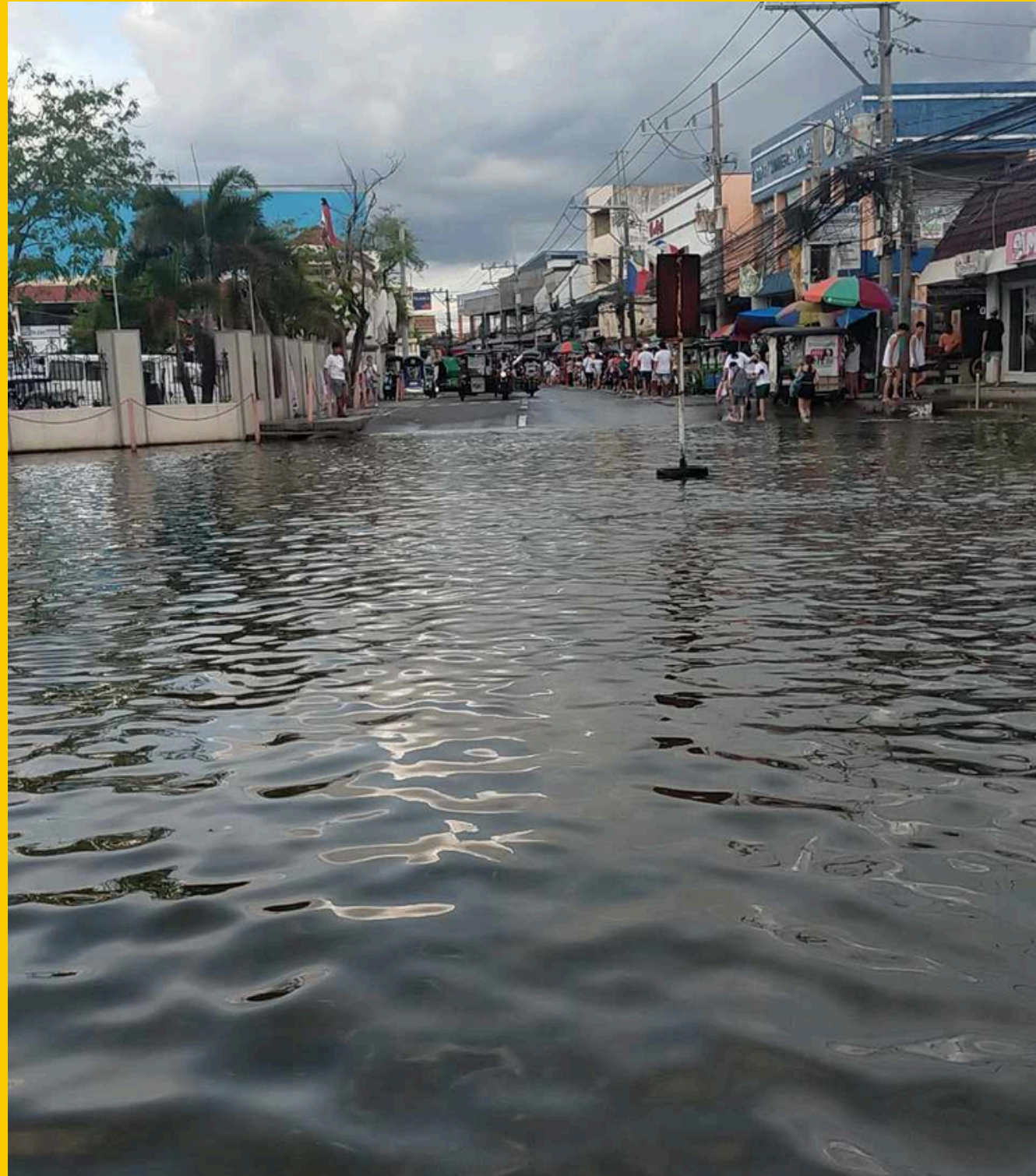
Effects of natural disasters

Communal Faucet
Systems / Stand
Posts

03

Impact on communities

Waterworks System
/ Individual House
Connections



Effects of Climate Change

Contributes to the rise of water-related diseases:

- Increased concentration of contaminants during droughts
- Increased contamination of water sources through flooding caused by storm surges

Climate change also leads to more **frequent, severe,** and **unpredictable extreme weather events**, which are often devastating enough to be classified as natural disasters.



Typhoon Haiyan (2013)

Thousands affected by cholera and other waterborne diseases, leading to widespread illness and fatalities

Typhoon Ketsana (2009)

Flooding caused severe outbreaks of leptospirosis and other infections

Effects of

Natural Disasters

- 💧 **Natural disasters** create a scarcity of safe, reliable water resources
- 💧 **Families** fall ill or die from water-related diseases
- 💧 **Diseases** spread through contaminated water consumption
- 💧 **Improper washing** of skin, ears, and eyes after contact with polluted water increases disease risk

Impact on Communities



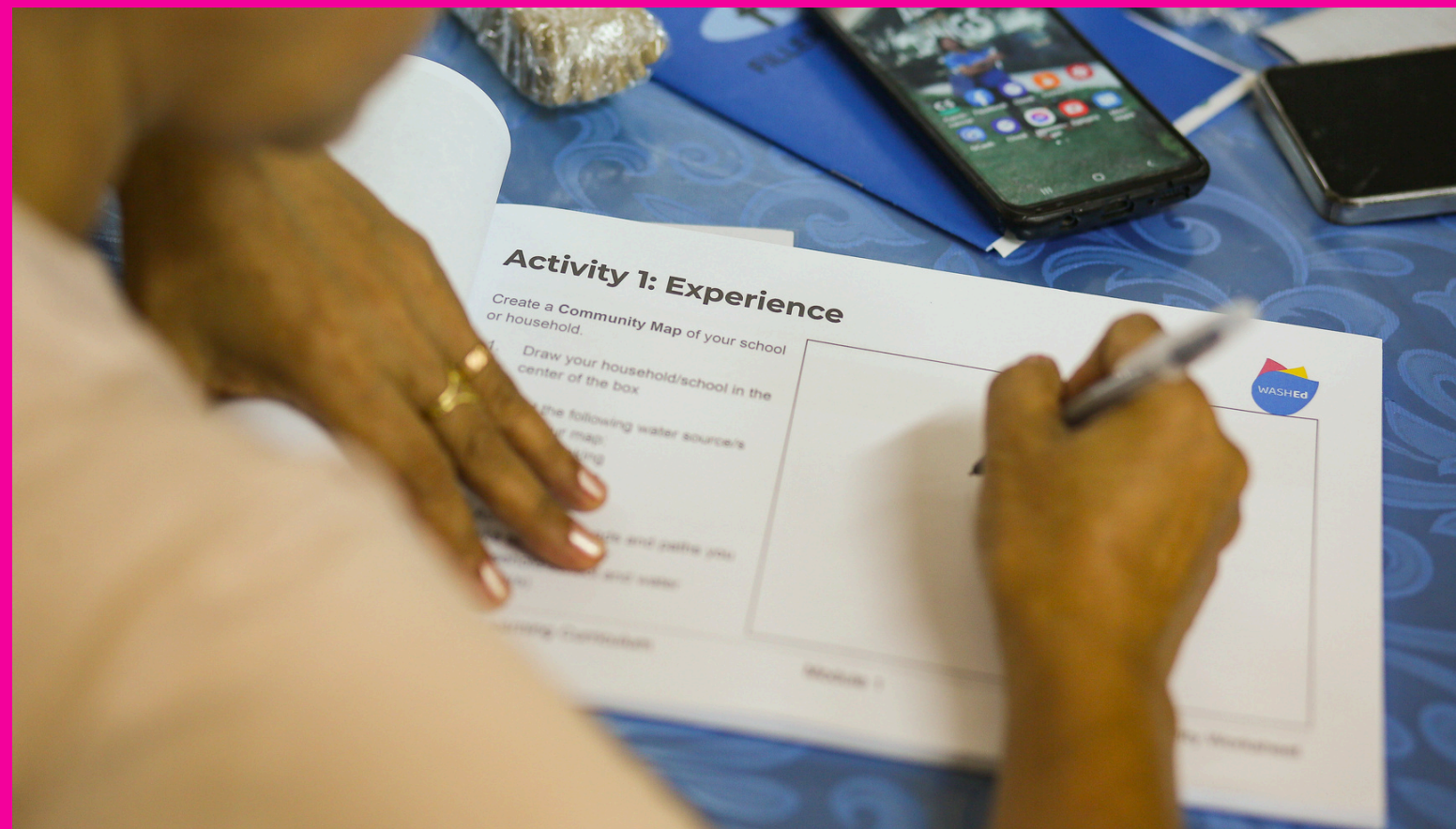
What can you do after a disaster?

- 1** Where no potable water is available, make sure to boil any untreated water, including tap water, before drinking
 - 2** Coordinate with your local government or healthcare workers if you start to experience any signs or symptoms associated with diseases
-  What are some other things you or your community can do to avoid water-related diseases?

Activities

to teach Module 2

- Classroom activities are designed using the **E-SPOT Framework** to engage Active Learning
- **Think-Pair-Share** approach is recommended for all activities



Activity 1: Experience

Activity 2: Share

Activity 3: Process & Observe

Activity 4: Try



Digital files for the classroom activities will be available from the WASH-Ed Learning Portal



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Congratulations

WASH
Education

You completed **Module 2: Water
Safety & Health**

JOIN

our community and
let's stay connected!



@washeducation



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